



兰州工业学院

实 验 报 告

(2023—2024 学年第 2 学期)

课程名称 《计算机网络应用》

专业班级 专升本电信 23

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实验项目	实验五 NAT 地址转换		
实验时间	2024 年 6 月 3 日	实验地点	信息与通信工程基础实验室 (EDA 技术实验室)
一、实验目的 1. 掌握配置 NAT 的 3 种方法； 2. 掌握静态 NAT 的配置方法； 二、实验环境 Cisco Packet Tracer 模拟软件 三、实验步骤 1、 按如图所示搭建实验环境，配置网络参数。拓扑图中的路由器设置时钟速率为 12800。 本实验要求通过对静态路由、静态 NAT，实现拓扑中全网的连通性。 图 1 实验拓扑结构图			

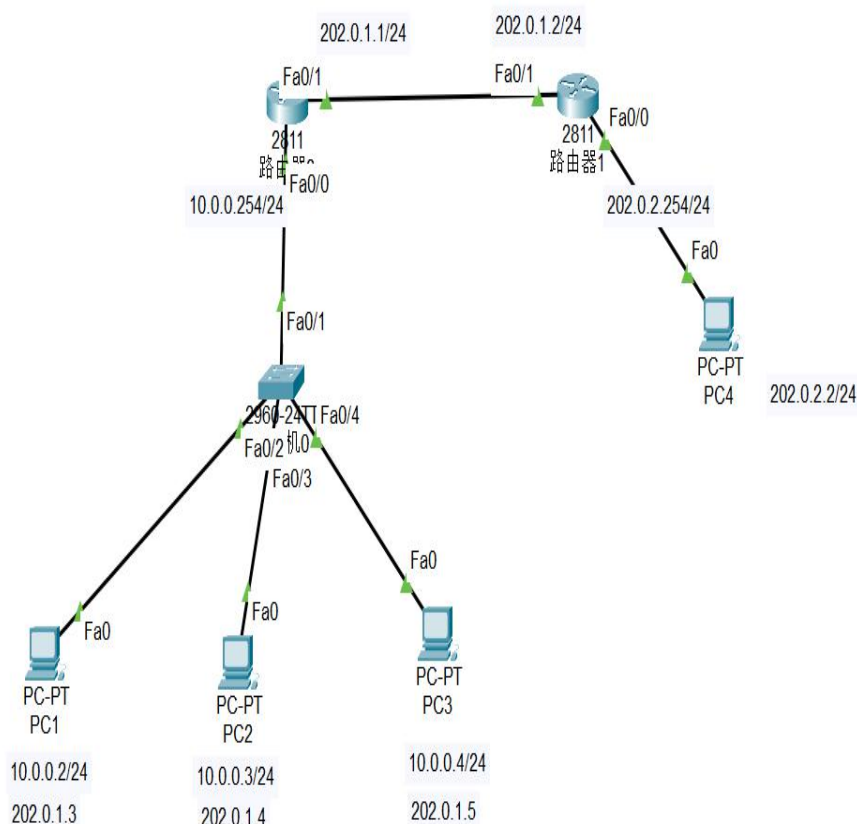


图 1 在 Cisco Packet Tracer 模拟软件的拓扑图

FastEthernet0/0

10.0.0.254/24

图 2 PC1、PC2（最上面两台计算机）连接到路由器相应端口的信息

路由器 0: FastEthernet0/0, 10.0.0.254/24

FastEthernet0/1, 202.0.1.1/24

路由器 1: FastEthernet0/0, 202.0.2.254/24

FastEthernet0/1, 202.0.1.2/24

图 3 路由器的端口信息（端口号及相应地址）

PC1: 10.0.0.2/24 10.0.0.254

PC2: 10.0.0.3/24 10.0.0.254

PC3: 10.0.0.4/24 10.0.0.254

图 4 PC1、PC2 的 IP 地址、子网掩码及网关信息

```

C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time=13ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms

C:\>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:

Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 202.0.2.2

Pinging 202.0.2.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 202.0.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

```

图 5 PC1 测试和其他 PC 的连通性

```

C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time=14ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 3ms

C:\>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:

Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 202.0.2.2

Pinging 202.0.2.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 202.0.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

图 6 PC2 测试和其他 PC 的连通性

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time=9ms TTL=128

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms

C:\>ping 202.0.2.2

Pinging 202.0.2.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 202.0.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

图 7 PC3 测试和其他 PC 的连通性

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

图 8 PC4 测试和其他 PC 的连通性

```

Router(config)#int fa0/0
Router(config-if)#ip nat ins
Router(config-if)#ip nat inside
Router(config-if)#exi
Router(config)#int fa0/1
Router(config-if)#ip nat outside
Router(config-if)#exi
Router(config)#ip nat inside source static 10.0.0.2 202.0.1.3
Router(config)#ip nat inside source static 10.0.0.3 202.0.1.4
Router(config)#ip nat inside source static 10.0.0.4 202.0.1.5
Router(config)#

```

复制

粘贴

图 9 路由器 0 的静态 NAT 命令

```

Router#show ip nat t
Pro Inside global      Inside local      Outside local      Outside global
icmp 202.0.1.3:1        10.0.0.2:1        202.0.2.2:1        202.0.2.2:1
icmp 202.0.1.3:2        10.0.0.2:2        202.0.2.2:2        202.0.2.2:2
icmp 202.0.1.3:3        10.0.0.2:3        202.0.2.2:3        202.0.2.2:3
icmp 202.0.1.3:4        10.0.0.2:4        202.0.2.2:4        202.0.2.2:4
icmp 202.0.1.3:5        10.0.0.2:5        202.0.2.2:5        202.0.2.2:5
icmp 202.0.1.3:6        10.0.0.2:6        202.0.2.2:6        202.0.2.2:6
icmp 202.0.1.3:7        10.0.0.2:7        202.0.2.2:7        202.0.2.2:7
icmp 202.0.1.3:8        10.0.0.2:8        202.0.2.2:8        202.0.2.2:8
--- 202.0.1.3          10.0.0.2          ---
--- 202.0.1.4          10.0.0.3          ---
--- 202.0.1.5          10.0.0.4          ---
Router#

```

复制

粘贴

图 10 PC 机私有 IP 地址映射为公共 IP 地址

```

Router(config-if)#ip route 202.0.2.0 255.255.255.0 202.0.1.2
Router(config)#ip route 202.0.2.0 255.255.255.0 202.0.1.2
Router(config)#int fa0/0
Router(config-if)#ip nat ins

```

图 11 路由器 0 的静态路由命令

```

Router(config)#ip route 10.0.0.0 255.255.255.0 202.0.1.1
Router(config)#

```

图 12 路由器 1 的静态路由命令

```

C:\>ping 10.0.0.3
Pinging 10.0.0.3 with 32 bytes of data:
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time=6ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms
C:\>ping 10.0.0.4
Pinging 10.0.0.4 with 32 bytes of data:
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 202.0.2.2
Pinging 202.0.2.2 with 32 bytes of data:
Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Ping statistics for 202.0.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>

```

图 13 PC1 与其他 PC 的连通性测试

```

C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:

Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128
Reply from 10.0.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 202.0.2.2

Pinging 202.0.2.2 with 32 bytes of data:

Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Reply from 202.0.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 202.0.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

图 14 PC2 与其他 PC 的连通性测试

```

C:\> ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128
Reply from 10.0.0.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\> ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128
Reply from 10.0.0.3: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\> ping 202.0.2.2

Pinging 202.0.2.2 with 32 bytes of data:

Reply from 202.0.2.2: bytes=32 time<1ms TTL=126
Reply from 202.0.2.2: bytes=32 time=1ms TTL=126
Reply from 202.0.2.2: bytes=32 time=7ms TTL=126
Reply from 202.0.2.2: bytes=32 time=1ms TTL=126

Ping statistics for 202.0.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 2ms

```

图 15 PC3 与其他 PC 的连通性测试

```
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 202.0.1.3: bytes=32 time<lms TTL=126
Reply from 202.0.1.3: bytes=32 time<lms TTL=126
Reply from 202.0.1.3: bytes=32 time<lms TTL=126
Reply from 202.0.1.3: bytes=32 time<lms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Reply from 202.0.1.4: bytes=32 time<lms TTL=126
Reply from 202.0.1.4: bytes=32 time<lms TTL=126
Reply from 202.0.1.4: bytes=32 time<lms TTL=126
Reply from 202.0.1.4: bytes=32 time<lms TTL=126

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:

Reply from 202.0.1.5: bytes=32 time<lms TTL=126
Reply from 202.0.1.5: bytes=32 time<lms TTL=126
Reply from 202.0.1.5: bytes=32 time<lms TTL=126
Reply from 202.0.1.5: bytes=32 time<lms TTL=126

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

图 16 PC4 与其他 PC 的连通性测试

四、总结

实验成绩评定表

序号	考核项目	分值分布	成绩
1	出勤与纪律情况	10	
2	实验任务完成情况	40	
3	实验报告质量	50	
总分			
指导教师签字			

备注：考核项目及分值分布应依据教学大纲确定。